作业

作业题目来自

梁昆淼《力学》 第 4 版 高等教育出版社

例： L01.08 梁昆淼书第 01 章第 08 题

Johns Analytical Mechanics 2nd ed Oxford University Press

例： J02.05 Johns 书第 02 章第 05 题

J03.07† 带 † 表示仅要求荣誉班同学完成

Homework 08, Week 12 2023.11.20 Due 2023.11.27 J19.01, J19.04, J19.05, J19.07, J19.10

★ J19.01

Consider a canonical transformation （正则变换）with constant coefficients, defined by

$$Q_i = \sum_{k=1}^{D} a_{ik}\, q_k \quad \text{or, in matrix form,} \quad [Q] = \mathbb{a}\,[q]$$

$$P_i = \sum_{k=1}^{D} b_{ik}\, p_k \quad \text{or, in matrix form,} \quad [P] = \mathbb{b}\,[p]$$

for $i = 1, 2, \ldots, D$, where the elements a_{ik} and b_{ik} of the matrices $\mathbb{a}$ and $\mathbb{b}$ are all constants.

(a) If the matrix $\mathbb{a}$ is given, find a general expression for the matrix $\mathbb{b}$.

(b) Canonical transformation of this type are of particular importance in quantum theory because the quantum substitution

$p_k \longrightarrow -i\hbar\,\partial/\partial q_k$ implies and is implied by the substitution $P_k \longrightarrow -i\hbar\,\partial/\partial Q_k$. Prove this implication.

(a) If the matrix $\mathbb{a}$ is

$\mathbb{a} = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix}$, find the numerical value of each element of the matrix $\mathbb{b}$

Solution

(a)

要作为正则变换，需要：$[Q_i,\ P_j] = \delta_{ij}$，故：$\sum_{k,l} a_{ik}\, b_{jl}\, [q_k,\ p_l] = \delta_{ij}$

即：$\sum_{k,l} a_{ik}\, b_{jl}\, \delta_{kl} = \delta_{ij} \implies \sum_{k,l} a_{ik}\, b_{jk} = \delta_{ij} \implies \mathbb{a}\,\mathbb{b}^T = I \implies \mathbb{b} = (\mathbb{a}^{-1})^T$

其它两个性质：$[Q_i,\ Q_j] = 0$, $[P_i,\ P_j] = 0$ 容易证明

(b)

若 $p_k \longrightarrow -i\hbar \dfrac{\partial}{\partial q_k}$，则 $P_k = -i\hbar \sum_l b_{kl} \dfrac{\partial}{\partial q_l}$

但 $\dfrac{\partial \varphi}{\partial q_l} = \sum_i \left[\dfrac{\partial \varphi}{\partial Q_i} \dfrac{\partial Q_i}{\partial q_l} + \dfrac{\partial \varphi}{\partial P_i} \dfrac{\partial P_i}{\partial q_l} \right] = \sum_i \dfrac{\partial \varphi}{\partial Q_i} a_{il}$

故：$P_k = -i\hbar \sum_i \sum_l b_{kl}\, a_{il} \dfrac{\partial}{\partial Q_i} = -i\hbar \sum_i \delta_{ik} \dfrac{\partial}{\partial Q_i} = -i\hbar \dfrac{\partial}{\partial Q_k}$

(c)

若 $\mathbb{a} = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix}$，则 $\mathbb{b} = (\mathbb{a}^{-1})^T = \dfrac{1}{10} \begin{pmatrix} 4 & -3 \\ 2 & 1 \end{pmatrix}$

★ J19.04

A transformation $q,\ p \longrightarrow Q,\ P$ is defined by

$$\begin{cases} Q_i = q_i \cos\theta_i + a_i\, p_i \sin\theta_i \\ P_i = -\dfrac{q_i}{a_i} \sin\theta_i + p_i \cos\theta_i \end{cases} \qquad i = 1,\ 2,\ \ldots,\ D$$

where θ_i, with $i = 1,\ 2,\ \ldots,\ D$, are independent, constant parameters, and a_i are given constant having appropriate units.

Use the Poisson-bracket condition to show that this transformation is canonical for any set of θ_i values.

Solution

$$\begin{aligned}
[Q_i,\ Q_j]_{\text{qp}} &= \sum_{k=1}^{D}\left(\frac{\partial Q_i}{\partial q_k}\frac{\partial Q_j}{\partial p_k} - \frac{\partial Q_i}{\partial p_k}\frac{\partial Q_j}{\partial q_k}\right) \\
&= \sum_{k=1}^{D}\left(\cos\theta_i\, \delta_{i\,k}\, a_j \sin\theta_j\, \delta_{k\,j} - a_i \sin\theta_i\, \delta_{k\,i} \cos\theta_j\, \delta_{k\,j}\right) \\
&= \left(\cos\theta_i\, a_j \sin\theta_j - a_i \sin\theta_i \cos\theta_j\right) \delta_{i\,j} = 0
\end{aligned}$$

$$\begin{aligned}
[P_i,\ P_j]_{\text{qp}} &= \sum_{k=1}^{D}\left(\frac{\partial P_i}{\partial q_k}\frac{\partial P_j}{\partial p_k} - \frac{\partial P_i}{\partial p_k}\frac{\partial P_j}{\partial q_k}\right) \\
&= \sum_{k=1}^{D}\left[-\frac{q_i}{a_i} \sin\theta_i\, \delta_{i\,k} \cos\theta_j\, \delta_{k\,j} - \cos\theta_i\, \delta_{k\,i}\left(-\frac{q_j}{a_j}\right) \sin\theta_j\, \delta_{k\,j}\right] \\
&= -\left(\frac{q_i}{a_i} \sin\theta_i \cos\theta_j - \frac{q_j}{a_j} \sin\theta_j \cos\theta_i\right) \delta_{i\,j} = 0
\end{aligned}$$

$$\begin{aligned}
[Q_i,\ P_j]_{\text{qp}} &= \sum_{k=1}^{D}\left(\frac{\partial Q_i}{\partial q_k}\frac{\partial P_j}{\partial p_k} - \frac{\partial Q_i}{\partial p_k}\frac{\partial P_j}{\partial q_k}\right) \\
&= \sum_{k=1}^{D}\left[\cos\theta_i\, \delta_{i\,k} \cos\theta_j\, \delta_{k\,j} - a_i \sin\theta_i\, \delta_{k\,i}\left(-\frac{1}{a_j}\right) \sin\theta_j\, \delta_{k\,j}\right] \\
&= \left(\cos\theta_i \cos\theta_j + \sin\theta_i \sin\theta_{i\,j}\right) \delta_{i\,j} = \delta_{i\,j}
\end{aligned}$$

所以，题中变换是正则变换

★ J19.05

Consider two point masses, m_1 and m_2, with position vectors $\vec{r}_i = x_i\, \vec{e}_x + y_i\, \vec{e}_y + z_i\, \vec{e}_z$, $(i = 1,\ 2)$.

Define the q variables to be $x_1,\ y_1,\ z_1,\ x_2,\ y_2,\ z_2$, the p variables to be $p_{1\,x},\ p_{1\,y},\ p_{1\,z},\ p_{2\,x},\ p_{2\,y},\ p_{2\,z}$

the Q variables to be $x,\ y,\ z,\ X,\ Y,\ Z$ and P variables to be $p_x,\ p_y,\ p_z,\ P_x,\ P_y,\ P_z$, where

$$\vec{r} = \vec{r}_2 - \vec{r}_1 = x\, \vec{e}_x + y\, \vec{e}_y + z\, \vec{e}_z\,, \quad \vec{R} = \frac{m_2\, \vec{r}_2 + m_1\, \vec{r}_1}{M} = X\, \vec{e}_x + Y\, \vec{e}_y + Z\, \vec{e}_z\,, \quad M = m_1 + m_2, \quad \mu = \frac{m_1\, m_2}{M}$$

$$\vec{p} = \mu\, \frac{d\,\vec{r}}{d\,t}, \quad \vec{P} = M\, \frac{d\,\vec{R}}{d\,t}, \qquad \vec{p}_i = m_i\, \frac{d\,\vec{r}_i}{d\,t} = p_{i\,x}\, \vec{e}_x + p_{i\,y}\, \vec{e}_y + p_{i\,z}\, \vec{e}_z, \quad i = 1,\ 2$$

(a) Prove that the transformation from $q,\ p$ to $Q,\ P$ is canonical

(*b*) Show that this is a canonical transformation of type $Q_i = \sum_{k=1}^{D} a_{i\,k}\, q_k$, $P_i = \sum_{k=1}^{D} b_{i\,k}\, q_k$, as discusse in Problem J19 .01 given above.

and that the standard quantum substitutions

$$\vec{p}_1 \longrightarrow -i\,\hbar\, \frac{\partial}{\partial \vec{r}_1}, \quad \vec{p}_2 \longrightarrow -i\,\hbar\, \frac{\partial}{\partial \vec{r}_2} \qquad \text{in the } q,\ p \text{ system imply the quntum substitutions}$$

$$\vec{p} \longrightarrow -i\,\hbar\, \frac{\partial}{\partial \vec{r}}, \quad \vec{P} \longrightarrow -i\,\hbar\, \frac{\partial}{\partial \vec{R}} \qquad \text{in the } Q,\ P \text{ system}$$

Solution

(*a*)

这是 J19 .01 的一个具体实例：$q=(\vec{r}_1,\vec{r}_2)$, $p=(\vec{p}_1,\vec{p}_2)$, $Q=(\vec{r},\vec{R})$, $P=(\vec{p},\vec{P})$,

雅克比矩阵 $J=\begin{pmatrix}\mathbb{a} & 0\\ 0 & \mathbb{b}\end{pmatrix}$,

$$\mathbb{a}=\frac{\partial Q}{\partial q}=\begin{pmatrix}-1 & 0 & 0 & 1 & 0 & 0\\ 0 & -1 & 0 & 0 & 1 & 0\\ 0 & 0 & -1 & 0 & 0 & 1\\ \frac{m_1}{M} & 0 & 0 & \frac{m_2}{M} & 0 & 0\\ 0 & \frac{m_1}{M} & 0 & 0 & \frac{m_2}{M} & 0\\ 0 & 0 & \frac{m_1}{M} & 0 & 0 & \frac{m_2}{M}\end{pmatrix},\quad \mathbb{b}=\frac{\partial P}{\partial p}=\begin{pmatrix}-\frac{m_2}{M} & 0 & 0 & \frac{m_1}{M} & 0 & 0\\ 0 & -\frac{m_2}{M} & 0 & 0 & \frac{m_1}{M} & 0\\ 0 & 0 & -\frac{m_2}{M} & 0 & 0 & \frac{m_1}{M}\\ 1 & 0 & 0 & 1 & 0 & 0\\ 0 & 1 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & 0 & 0 & 1\end{pmatrix}$$

故：$\mathbb{a}\,\mathbb{b}^T=U$

$$J\,s\,J^T=\begin{pmatrix}\mathbb{a} & 0\\ 0 & \mathbb{b}\end{pmatrix}\begin{pmatrix}0 & U\\ -U & 0\end{pmatrix}\begin{pmatrix}\mathbb{a}^T & 0\\ 0 & \mathbb{b}^T\end{pmatrix}=\begin{pmatrix}0 & \mathbb{a}\\ -\mathbb{b} & 0\end{pmatrix}\begin{pmatrix}\mathbb{a}^T & 0\\ 0 & \mathbb{b}^T\end{pmatrix}=\begin{pmatrix}0 & \mathbb{a}\,\mathbb{b}^T\\ \mathbb{a}^T\mathbb{b} & 0\end{pmatrix}=\begin{pmatrix}0 & U\\ -U & 0\end{pmatrix}=s$$ 是正则变换

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a = {{-1, 0, 0, 1, 0, 0}, {0, -1, 0, 0, 1, 0}, {0, 0, -1, 0, 0, 1},
    {m1/M, 0, 0, m2/M, 0, 0}, {0, m1/M, 0, 0, m2/M, 0}, {0, 0, m1/M, 0, 0, m2/M}};
b = {{-m2/M, 0, 0, m1/M, 0, 0}, {0, -m2/M, 0, 0, m1/M, 0}, {0, 0, -m2/M, 0, 0, m1/M},
    {1, 0, 0, 1, 0, 0}, {0, 1, 0, 0, 1, 0}, {0, 0, 1, 0, 0, 1}};
bt = Transpose[b];
Simplify[a.bt /. M → m1 + m2]
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{{1, 0, 0, 0, 0, 0}, {0, 1, 0, 0, 0, 0}, {0, 0, 1, 0, 0, 0},
 {0, 0, 0, 1, 0, 0}, {0, 0, 0, 0, 1, 0}, {0, 0, 0, 0, 0, 1}}
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(*b*)

显然，$Q_i=\sum_{k=1}^{D}a_{ik}\,q_k$, $P_i=\sum_{k=1}^{D}b_{ik}\,q_k$, 且 $\mathbb{a}\mathbb{b}^T=U$

由上题：$$\begin{cases}\vec{p}_1\longrightarrow -i\hbar\dfrac{\partial}{\partial\vec{r}_1}\\[2ex] \vec{p}_2\longrightarrow -i\hbar\dfrac{\partial}{\partial\vec{r}_2}\end{cases}\Longrightarrow\begin{cases}\vec{p}\longrightarrow -i\hbar\dfrac{\partial}{\partial\vec{r}}\\[2ex] \vec{P}\longrightarrow -i\hbar\dfrac{\partial}{\partial\vec{R}}\end{cases}$$

★ J19.07

We know that each product $q_k\,p_k$ for $k=1,2,\ldots,D$ has units of action, which are $M\,L^2/T$.

Demonstrate from the Poisson bracket condition (long form) of canonical transformation that, for all $k=1,2,\ldots,D$, each product $Q_k\,P_k$ after any canonical transformation will have the units of action.

Solution

泊松括号：$[Q_i,P_j]_{\text{qp}}=\delta_{ij}$, 而 $[Q_i,P_j]_{\text{qp}}=\sum_{k=1}^{D}\left(\frac{\partial Q_i}{\partial q_k}\frac{\partial P_j}{\partial p_k}-\frac{\partial Q_i}{\partial p_k}\frac{\partial P_j}{\partial q_k}\right)$

分母是 $q_k\,p_k$，是作用量量纲，故分子也应该是作用量量纲，以保证泊松括号的结果 δ_{ij} 是无量纲的

本题的结论是：任何正则变换得到的 $Q_k\,P_k$，均有作用量的量纲。

★ J19.10

A system with one degree of freedom has a Hamiltonian given by

$$H(q,\ p,\ t) = \frac{a^2}{2\,m\,p^2} + \frac{m\,\omega^2\,q^2\,p^4}{2\,a^2}$$

(*a*) Consider the transformation $q,\ p \longrightarrow Q,\ P$ defined as $Q = b\,q\,p^2,\quad P = -\dfrac{a}{p}$

Assuming the constant *a* given, for what value of *b* is this transformation canonical?

(*b*) Use this transformation with your chosen value of *b* to write the Hamiltonian in terms of the $Q,\ P$ variables.

(*c*) Solve the Hamiltonian obtained in part (*b*) and use the inverse transformation to get the system trajectory, *q* as a function of *t*.

What do your learn from this problem?

Solution

(*a*)

$$J = \begin{pmatrix} b\,p^2 & 2\,b\,q\,p \\ 0 & \dfrac{a}{p^2} \end{pmatrix} \quad \Longrightarrow$$

$$J\,s\,J^T = \begin{pmatrix} b\,p^2 & 2\,b\,q\,p \\ 0 & \dfrac{a}{p^2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} b\,p^2 & 0 \\ 2\,b\,q\,p & \dfrac{a}{p^2} \end{pmatrix} = \begin{pmatrix} -2\,b\,q\,p & b\,p^2 \\ -\dfrac{a}{p^2} & 0 \end{pmatrix} \begin{pmatrix} b\,p^2 & 0 \\ 2\,b\,q\,p & \dfrac{a}{p^2} \end{pmatrix} = \begin{pmatrix} 0 & a\,b \\ -a\,b & 0 \end{pmatrix}$$

$b = a^{-1}$ 时构成正则变换

另解：$[Q,\ P]_{qp} = 1 \quad \Longrightarrow \quad \left[b\,q\,p^2,\ -\dfrac{a}{p}\right]_{qp} = -a\,b\,p^2\left[q,\ \dfrac{1}{p}\right] = -a\,b\,p^2\left(-\dfrac{1}{p^2}\right)[q,\ p] = a\,b\ =\ 1$

(*b*)

$$\begin{cases} Q = \dfrac{1}{a}\,q\,p^2 \\ P = -\dfrac{a}{p} \end{cases} \Longrightarrow \begin{cases} q = \dfrac{Q\,P^2}{a} \\ p = -\dfrac{a}{P} \end{cases} \Longrightarrow H = \frac{P^2}{2\,m} + \frac{m\,\omega^2}{2}\,Q^2$$

(*c*)

$$H = \frac{P^2}{2\,m} + \frac{m\,\omega^2}{2}\,Q^2 \Longrightarrow \begin{cases} \dot{Q} = \dfrac{\partial H}{\partial P} = \dfrac{P}{m} \\ \dot{P} = -\dfrac{\partial H}{\partial Q} = -m\,\omega^2\,Q \end{cases} \Longrightarrow \ddot{Q} + \omega^2\,Q = 0 \quad \Longrightarrow \begin{cases} Q = Q_0\cos(\omega\,t + \phi) \\ P = m\,\dot{Q} = -m\,\omega\,Q_0\sin(\omega\,t + \phi) \end{cases}$$

$$\Longrightarrow \quad q = \frac{Q\,P^2}{a} = \frac{1}{a}\,m^2\,\omega^2\,Q_0^3\cos(\omega\,t + \phi)\sin^2(\omega\,t + \phi)$$

由此题可知，可以通过正则变换简化并求解运动方程